

ARCHIT AGRAWAL

6233120435 | architagrawal000@gmail.com | www.linkedin.com/in/agrawal-archit | github.com/architagrawal | <https://agrawal-archit.vercel.app>

PROFESSIONAL EXPERIENCE

AI Engineer

03/2025 - Present

MyStage Music Inc.

- Automated **large-scale event metadata processing workflows handling 70K+ records/day** by orchestrating extraction, enrichment, validation, and deduplication services across distributed Python pipelines
- Built **context-aware agent workflows using LangGraph** to coordinate multi-step decision execution, tool routing, retries, and structured validation for noisy real-world datasets
- Designed **retrieval and reasoning architecture combining vector search, graph traversal, and structured metadata indexing** to improve entity resolution accuracy across fragmented data sources
- Developed **MCP-style tool interfaces** enabling LLM-driven services to execute APIs, retrieve operational context, and coordinate cross-system workflow execution
- Replaced **manual media processing workflows** with multimodal OCR + LLM extraction pipelines that converted unstructured images and documents into structured searchable records
- Implemented **schema-governed execution layers using Pydantic AI** to enforce deterministic outputs, validation rules, and safe downstream automation behavior
- Built **tracing and evaluation infrastructure for production LLM workflows**, enabling workflow auditing, failure analysis, and debugging of agent reasoning paths
- Deployed **containerized Python services using Docker, GitHub Actions, and GCP infrastructure** with automated CI/CD pipelines and scalable runtime orchestration

Software Developer

09/2023 - 05/2025

Ed-Plus, Arizona State University

- Built **production RAG platform serving 6,000+ users** by combining Neo4j knowledge graphs, semantic retrieval, and structured metadata pipelines for grounded LLM responses
- Automated **academic document ingestion and validation workflows** using Python data pipelines processing 10,000+ transcript and knowledge records
- Developed **async FastAPI inference services with semantic caching and retrieval orchestration**, improving response latency and throughput under concurrent workloads
- Designed **evaluation and validation frameworks comparing LLM outputs against structured ground-truth datasets** to improve response consistency and reduce invalid generations
- Created **internal tooling for prompt evaluation, retrieval experimentation, and agent workflow testing** across multiple LLM configurations
- Implemented **structured fallback and validation logic** preventing malformed or hallucinated outputs from reaching production APIs

Software Developer Intern

05/2024 - 08/2024

Knowledge Exchange for Resilience, Arizona State University

- Automated **high-volume ETL workflows using Python and PostgreSQL** processing 10,000+ daily faculty records with validation layers improving downstream data reliability
- Developed and deployed **20+ production REST APIs** using .NET Core and AWS EC2 infrastructure
- Reduced **API response latency from 198ms to 20ms** through Redis caching and SQL query optimization strategies
- Built **automated load-testing and validation framework using Locust** to simulate production-scale traffic and identify operational bottlenecks before deployment

Software Engineer

01/2022 - 08/2023

Zeus Learning

- Migrated **monolithic applications into Docker and Kubernetes microservices architecture** supporting 10,000+ daily enterprise requests with high-availability deployments
- Built **distributed backend workflows using RabbitMQ** for asynchronous processing, service decoupling, and fault-tolerant task execution
- Automated **deployment validation and release workflows using GitLab CI and Jenkins pipelines**, reducing deployment time by 70%
- Improved **production reliability** by implementing automated CI/CD checks, rollback mechanisms, and infrastructure monitoring workflows
- Developed **ML-driven occupancy prediction platform for 300+ enterprise locations**, improving workplace utilization rates by 30%

Software Developer Intern

09/2021 - 12/2021

EAT.FIT

- Built **real-time order tracking workflows** using React, Express, and WebSockets, improving live delivery visibility and operational reliability
- Reduced **external API usage costs by 45%** through optimization of Google Maps API request orchestration and caching strategies

TECHNICAL SKILLS

Languages: Python, Go, C#, JavaScript, TypeScript, SQL.

LLM & Agentic AI: Hugging Face (Transformers, Diffusers), LangGraph, CrewAI, Pydantic AI, MCP, LangChain, LlamaIndex, RAG Pipelines, Vector Stores (Pinecone/FAISS), OpenAI API, OCR (Tesseract/PaddleOCR), Voice Generation (Bark), Video Generation (SVD).

HPC & Compute: CUDA, C++, Linux (Bash/Shell), GPU Profiling (Nsight), Docker, Kubernetes, AWS (EC2, Lambda), GCP (Vertex AI).

Web & Backend: FastAPI, Django, Flask, .NET Core, Node.js, React.js, GraphQL.

Data & Observability: SQLAlchemy, PyTorch, TensorFlow, NumPy, Pandas, Logfire (LLM Tracing), Locust, Redis, RabbitMQ.

Databases: PostgreSQL, MySQL, Redis, Neo4j (Graph DB), Pinecone (Vector DB), DynamoDB

Cloud & DevOps: AWS (EC2, Lambda, EKS, SQS, S3), Google Cloud Platform, Docker, Kubernetes, Jenkins, GitLab, Nginx, RabbitMQ

EDUCATION

Master of Science, Computer Science	08/2023 - 05/2025
Arizona State University (ASU)	GPA: 4.0
Bachelor of Technology, Information and Communication Technology	08/2018 - 05/2022
Dhirubhai Ambani Institute of Information and Communication Technology	

PROJECTS

PrismSplit Expense Management Application	12/2025 - Present
- Developing a mobile application with offline sync for granular bill splitting, allowing users to parse receipts and assign specific items to individuals rather than simple even splits. (React Native, Expo, Zustand, TypeScript, TanStack, Tamagui) - Engineered the backend logic to handle many-to-many relationships between users, receipts, and line items, ensuring data consistency across shared groups. (Supabase, SQL)	
SRP Electric MCP Server	12/2025 - 1/2025
- Built a TypeScript MCP server to facilitate automated, agentic AI interactions with energy systems. - Operated in a highly ambiguous environment to reverse-engineer the undocumented authentication flow of a legacy utility portal, converting unstructured enterprise data into structured JSON tools for secure AI reasoning.	
Auto-Scaling Image Recognition as a Service	1/2024 - 04/2024
Constructed an AWS cloud infrastructure SaaS using AWS EC2, AWS SQS, Lambda, API gateway, Terraform and amplify for auto linear scaling based on demand providing image recognition service through deep learning models, serving 100 concurrent requests in 5 seconds. (AWS SDK)	
MCP-Based GitHub PR Review Automation Agent	07/2025 - 08/2025
- Built MCP-based workflow automation service integrating GitHub webhooks, LLM reasoning, and automated pull request analysis pipelines - Designed context-routing logic enabling agents to retrieve repository state, ticket metadata, and CI/CD execution context before generating review decisions - Automated engineering workflows across GitHub Actions and Asana to update statuses, link tickets, and generate review summaries without manual coordination	